EEOB 546 R Assignment

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Remove objects you've created and load required libraries before running my code using this chunk:

```
rm(list = ls())
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.1.4     v readr     2.2.0
## v forcats   1.0.1     v stringr   1.6.0
## v ggplot2   4.0.1     v tibble    3.3.1
## v lubridate 1.9.5     v tidyr     1.3.2
## v purrr     1.2.1
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

Part I

1. Data Inspection

```
# load the dataframes
fang <- read_tsv ("https://raw.githubusercontent.com/EEOB-BioData/BCB5460_Spring2026/refs/heads/main/as
```

```
## Rows: 2782 Columns: 986
## -- Column specification --------------------------------------------------------
## Delimiter: "\t"
## chr (986): Sample_ID, JG_OTU, Group, abph1.20, abph1.22, ae1.3, ae1.4, ae1.5...
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
snp <- read_tsv("https://raw.githubusercontent.com/EEOB-BioData/BCB5460_Spring2026/refs/heads/main/assi
```

```
## Rows: 983 Columns: 15
## -- Column specification --------------------------------------------------------
## Delimiter: "\t"
```

```
## chr (9): SNP_ID, Chromosome, Position, alt_pos, mult_positions, amplicon, cd...
## dbl (6): cdv_marker_id, Genaissance_daa_id, Sequenom_daa_id, count_amplicons...
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```r
head(fang)
```

```
## # A tibble: 6 x 986
##   Sample_ID JG_OTU   Group abph1.20 abph1.22 ae1.3 ae1.4 ae1.5 an1.4 ba1.6 ba1.9
##   <chr>     <chr>    <chr> <chr>    <chr>    <chr> <chr> <chr> <chr> <chr> <chr>
## 1 SL-15     T-aust-1 TRIPS ?/?      ?/?      T/T   G/G   T/T   C/C   ?/?   G/G
## 2 SL-16     T-aust-2 TRIPS ?/?      ?/?      T/T   ?/?   T/T   C/C   A/G   G/G
## 3 SL-11     T-brav-1 TRIPS ?/?      ?/?      T/T   G/G   T/T   ?/?   G/G   G/G
## 4 SL-12     T-brav-2 TRIPS ?/?      ?/?      T/T   G/G   T/T   C/C   G/G   G/G
## 5 SL-18     T-cund   TRIPS ?/?      ?/?      T/T   G/G   T/T   C/C   ?/?   G/G
## 6 SL-2      T-dact-1 TRIPS ?/?      ?/?      T/T   G/G   T/T   C/C   A/G   G/G
## # i 975 more variables: bt2.5 <chr>, bt2.7 <chr>, bt2.8 <chr>, Fea2.1 <chr>,
## #   Fea2.5 <chr>, id1.3 <chr>, lg2.11 <chr>, lg2.2 <chr>, pbf1.1 <chr>,
## #   pbf1.2 <chr>, pbf1.3 <chr>, pbf1.5 <chr>, pbf1.6 <chr>, pbf1.7 <chr>,
## #   pbf1.8 <chr>, PZA00003.11 <chr>, PZA00004.2 <chr>, PZA00005.8 <chr>,
## #   PZA00005.9 <chr>, PZA00006.13 <chr>, PZA00006.14 <chr>, PZA00008.1 <chr>,
## #   PZA00010.5 <chr>, PZA00013.10 <chr>, PZA00013.11 <chr>, PZA00013.9 <chr>,
## #   PZA00015.4 <chr>, PZA00017.1 <chr>, PZA00018.5 <chr>, ...
```

```r
ncol(fang)
```

```
## [1] 986
```

```r
nrow(fang)
```

```
## [1] 2782
```

```r
dim(fang)
```

```
## [1] 2782  986
```

```r
colnames(fang)
```

```
##   [1] "Sample_ID"      "JG_OTU"         "Group"          "abph1.20"
##   [5] "abph1.22"       "ae1.3"          "ae1.4"          "ae1.5"
##   [9] "an1.4"          "ba1.6"          "ba1.9"          "bt2.5"
##  [13] "bt2.7"          "bt2.8"          "Fea2.1"         "Fea2.5"
##  [17] "id1.3"          "lg2.11"         "lg2.2"          "pbf1.1"
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## [657] "PZA03089.12"    "PZA03090.31"    "PZA03092.7"     "PZA03094.18"
## [661] "PZA03094.6"     "PZA03095.1"     "PZA03095.2"     "PZA03095.3"
## [665] "PZA03097.4"     "PZA03097.7"     "PZA03097.9"     "PZA03102.10"
## [669] "PZA03102.2"     "PZA03102.9"     "PZA03137.1"     "PZA03172.2"
## [673] "PZA03223.3"     "PZA03258.2"     "PZA03283.2"     "PZA03284.3"
## [677] "PZA03290.1"     "PZA03290.2"     "PZA03295.4"     "PZA03296.6"
## [681] "PZA03296.7"     "PZA03298.1"     "PZA03298.2"     "PZA03301.2"
## [685] "PZA03301.4"     "PZA03302.1"     "PZA03305.6"     "PZA03305.7"
## [689] "PZA03311.2"     "PZA03311.3"     "PZA03311.4"     "PZA03311.5"
```

```
## [693] "PZA03312.1"     "PZA03312.2"     "PZA03316.2"     "PZA03317.1"
## [697] "PZA03319.3"     "PZA03319.4"     "PZA03320.3"     "PZA03320.4"
## [701] "PZA03328.5"     "PZA03329.1"     "PZA03329.2"     "PZA03333.3"
## [705] "PZA03335.2"     "PZA03335.3"     "PZA03337.1"     "PZA03338.5"
## [709] "PZA03340.2"     "PZA03342.2"     "PZA03344.4"     "PZA03344.5"
## [713] "PZA03344.6"     "PZA03345.1"     "PZA03345.2"     "PZA03345.4"
## [717] "PZA03347.1"     "PZA03348.1"     "PZA03349.1"     "PZA03349.9"
## [721] "PZA03767.1"     "PZA03767.4"     "PZA03767.5"     "PZA03773.2"
## [725] "PZA03773.3"     "PZA03774.1"     "PZA03774.10"    "PZA03774.2"
## [729] "PZA03774.4"     "PZA03774.5"     "PZA03774.6"     "PZA03774.8"
## [733] "PZA03774.9"     "PZA03775.1"     "PZA03775.11"    "PZA03775.2"
## [737] "PZA03775.3"     "PZA03775.4"     "PZA03775.6"     "PZA03775.7"
## [741] "PZA03775.8"     "PZA03775.9"     "PZA03781.1"     "PZA03781.2"
## [745] "PZA03781.3"     "PZA03781.4"     "PZA03781.5"     "PZA03781.6"
## [749] "PZA03781.7"     "PZA03781.8"     "PZA03782.1"     "PZA03782.3"
## [753] "PZA03786.1"     "PZA03786.2"     "PZA03789.1"     "PZA03789.2"
## [757] "PZA03789.4"     "PZB00011.4"     "PZB00011.5"     "PZB00041.2"
## [761] "PZB00041.4"     "PZB00049.2"     "PZB00049.4"     "PZB00049.7"
## [765] "PZB00055.1"     "PZB00060.4"     "PZB00062.6"     "PZB00062.7"
## [769] "PZB00062.8"     "PZB00067.2"     "PZB00067.3"     "PZB00067.4"
## [773] "PZB00067.5"     "PZB00078.1"     "PZB00081.2"     "PZB00081.4"
## [777] "PZB00081.5"     "PZB00081.7"     "PZB00092.1"     "PZB00092.4"
## [781] "PZB00093.3"     "PZB00093.4"     "PZB00093.6"     "PZB00096.2"
## [785] "PZB00096.3"     "PZB00136.3"     "PZB00140.1"     "PZB00145.2"
## [789] "PZB00149.2"     "PZB00149.4"     "PZB00153.1"     "PZB00153.2"
## [793] "PZB00153.3"     "PZB00153.5"     "PZB00160.1"     "PZB00160.2"
## [797] "PZB00160.4"     "PZB00165.2"     "PZB00165.6"     "PZB00169.4"
## [801] "PZB00169.6"     "PZB00175.1"     "PZB00175.2"     "PZB00175.3"
## [805] "PZB00175.4"     "PZB00175.5"     "PZB00180.1"     "PZB00180.2"
## [809] "PZB00183.3"     "PZB00188.6"     "PZB00207.3"     "PZB00221.3"
## [813] "PZB00221.8"     "PZB00229.3"     "PZB00232.1"     "PZB00232.2"
## [817] "PZB00232.4"     "PZB00232.5"     "PZB00379.3"     "PZB00379.4"
## [821] "PZB00379.5"     "PZB00393.7"     "PZB00409.3"     "PZB00416.2"
## [825] "PZB00416.5"     "PZB00454.2"     "PZB00454.3"     "PZB00454.4"
## [829] "PZB00454.5"     "PZB00498.2"     "PZB00498.4"     "PZB00598.1"
## [833] "PZB00598.2"     "PZB00603.3"     "PZB00603.4"     "PZB00603.5"
## [837] "PZB00607.2"     "PZB00761.1"     "PZB00761.2"     "PZB00849.2"
## [841] "PZB00849.3"     "PZB00849.4"     "PZB00859.1"     "PZB01109.2"
## [845] "PZB01109.3"     "PZB01110.1"     "PZB01110.2"     "PZB01110.3"
## [849] "PZB01111.6"     "PZB01111.7"     "PZB01111.8"     "PZB01112.3"
## [853] "PZB01112.4"     "PZB01112.5"     "PZB01112.6"     "PZB01113.4"
## [857] "PZB01114.1"     "PZB01114.3"     "PZB01115.1"     "PZB01115.5"
## [861] "PZB01115.6"     "PZB01116.2"     "PZB01221.1"     "PZB01222.1"
## [865] "PZB01222.3"     "PZB01223.3"     "PZB01223.4"     "PZB01223.7"
## [869] "PZB01225.1"     "PZB01225.2"     "PZB01225.4"     "PZB01228.1"
## [873] "PZB01228.3"     "PZB01228.4"     "PZB01233.2"     "PZB01233.3"
## [877] "PZB01238.5"     "PZB01238.6"     "PZB01427.1"     "PZB01427.3"
## [881] "PZB01463.2"     "PZB01463.3"     "PZB01463.4"     "PZD00003.1"
## [885] "PZD00003.3"     "PZD00007.1"     "PZD00008.3"     "PZD00011.1"
## [889] "PZD00011.3"     "PZD00011.4"     "PZD00012.1"     "PZD00012.2"
## [893] "PZD00012.3"     "PZD00012.4"     "PZD00012.5"     "PZD00013.3"
## [897] "PZD00013.4"     "PZD00014.3"     "PZD00017.1"     "PZD00019.1"
## [901] "PZD00020.2"     "PZD00020.3"     "PZD00020.4"     "PZD00020.6"
## [905] "PZD00021.2"     "PZD00021.4"     "PZD00021.5"     "PZD00022.1"
```

```
## [909] "PZD00022.3"     "PZD00022.4"     "PZD00024.2"     "PZD00025.1"
## [913] "PZD00025.2"     "PZD00030.1"     "PZD00030.4"     "PZD00030.5"
## [917] "PZD00030.6"     "PZD00034.3"     "PZD00043.1"     "PZD00043.2"
## [921] "PZD00043.3"     "PZD00043.4"     "PZD00044.2"     "PZD00044.3"
## [925] "PZD00044.4"     "PZD00045.1"     "PZD00045.2"     "PZD00045.3"
## [929] "PZD00045.4"     "PZD00049.3"     "PZD00049.4"     "PZD00049.5"
## [933] "PZD00051.1"     "PZD00052.3"     "PZD00052.4"     "PZD00062.2"
## [937] "PZD00066.1"     "PZD00067.1"     "PZD00067.2"     "PZD00067.3"
## [941] "PZD00068.1"     "PZD00069.2"     "PZD00069.3"     "PZD00069.4"
## [945] "PZD00069.5"     "PZD00073.1"     "PZD00073.2"     "PZD00073.6"
## [949] "PZD00074.1"     "PZD00075.1"     "PZD00075.2"     "PZD00076.1"
## [953] "PZD00076.2"     "PZD00076.4"     "PZD00077.10"    "PZD00077.5"
## [957] "PZD00077.7"     "PZD00077.8"     "PZD00078.2"     "Ra2_ORF.1"
## [961] "Ra2_ORF.2"      "Ra2_ORF.4"      "Ra2_promoter.1" "Ra2_promoter.2"
## [965] "Ra2_promoter.3" "sh2.5"          "sh2.6"          "sh2.7"
## [969] "sh2.9"          "su1.4"          "su1.5"          "su1.7"
## [973] "tb1.17"         "tb1.18"         "tb1.19"         "tb1.5"
## [977] "te1.3"          "te1.4"          "zagl1.1"        "zagl1.6"
## [981] "zap1.2"         "zen1.1"         "zen1.2"         "zen1.4"
## [985] "zfl2.6"         "zmm3.4"
```

```r
str(fang)
```

```
## spc_tbl_ [2,782 x 986] (S3: spec_tbl_df/tbl_df/tbl/data.frame)
##  $ Sample_ID     : chr [1:2782] "SL-15" "SL-16" "SL-11" "SL-12" ...
##  $ JG_OTU        : chr [1:2782] "T-aust-1" "T-aust-2" "T-brav-1" "T-brav-2" ...
##  $ Group         : chr [1:2782] "TRIPS" "TRIPS" "TRIPS" "TRIPS" ...
##  $ abph1.20      : chr [1:2782] "?/?" "?/?" "?/?" "?/?" ...
##  $ abph1.22      : chr [1:2782] "?/?" "?/?" "?/?" "?/?" ...
##  $ ae1.3         : chr [1:2782] "T/T" "T/T" "T/T" "T/T" ...
##  $ ae1.4         : chr [1:2782] "G/G" "?/?" "G/G" "G/G" ...
##  $ ae1.5         : chr [1:2782] "T/T" "T/T" "T/T" "T/T" ...
##  $ an1.4         : chr [1:2782] "C/C" "C/C" "?/?" "C/C" ...
##  $ ba1.6         : chr [1:2782] "?/?" "A/G" "G/G" "G/G" ...
##  $ ba1.9         : chr [1:2782] "G/G" "G/G" "G/G" "G/G" ...
##  $ bt2.5         : chr [1:2782] "?/?" "?/?" "C/C" "C/C" ...
##  $ bt2.7         : chr [1:2782] "A/A" "A/A" "A/A" "A/A" ...
##  $ bt2.8         : chr [1:2782] "?/?" "?/?" "?/?" "?/?" ...
##  $ Fea2.1        : chr [1:2782] "C/C" "C/C" "?/?" "?/?" ...
##  $ Fea2.5        : chr [1:2782] "A/A" "A/A" "A/A" "A/A" ...
##  $ id1.3         : chr [1:2782] "T/T" "T/T" "T/T" "T/T" ...
##  $ lg2.11        : chr [1:2782] "C/C" "C/C" "C/C" "C/C" ...
##  $ lg2.2         : chr [1:2782] "A/A" "A/A" "A/A" "A/A" ...
##  $ pbf1.1        : chr [1:2782] "?/?" "T/T" "T/T" "T/T" ...
##  $ pbf1.2        : chr [1:2782] "?/?" "?/?" "?/?" "?/?" ...
##  $ pbf1.3        : chr [1:2782] "?/?" "?/?" "?/?" "?/?" ...
##  $ pbf1.5        : chr [1:2782] "?/?" "?/?" "A/A" "A/A" ...
##  $ pbf1.6        : chr [1:2782] "?/?" "?/?" "?/?" "?/?" ...
##  $ pbf1.7        : chr [1:2782] "C/C" "C/C" "C/C" "C/C" ...
##  $ pbf1.8        : chr [1:2782] "C/C" "C/C" "C/C" "C/C" ...
##  $ PZA00003.11   : chr [1:2782] "?/?" "?/?" "C/C" "?/?" ...
##  $ PZA00004.2    : chr [1:2782] "T/T" "T/T" "?/?" "T/T" ...
##  $ PZA00005.8    : chr [1:2782] "G/G" "G/G" "G/G" "G/G" ...
##  $ PZA00005.9    : chr [1:2782] "C/C" "C/C" "C/C" "C/C" ...
```

```
##  $ PZA00006.13   : chr [1:2782] "A/A" "A/A" "A/A" "A/A" ...
##  $ PZA00006.14   : chr [1:2782] "?/?" "G/G" "G/G" "G/G" ...
##  $ PZA00008.1    : chr [1:2782] "C/C" "C/C" "C/C" "C/C" ...
##  $ PZA00010.5    : chr [1:2782] "C/C" "C/C" "C/C" "C/C" ...
##  $ PZA00013.10   : chr [1:2782] "A/A" "A/A" "A/A" "A/A" ...
##  $ PZA00013.11   : chr [1:2782] "C/C" "C/C" "C/T" "C/T" ...
##  $ PZA00013.9    : chr [1:2782] "T/T" "T/T" "T/T" "T/T" ...
##  $ PZA00015.4    : chr [1:2782] "?/?" "?/?" "?/?" "?/?" ...
##  $ PZA00017.1    : chr [1:2782] "G/G" "G/G" "G/G" "G/G" ...
##  $ PZA00018.5    : chr [1:2782] "C/C" "C/C" "C/C" "C/C" ...
##  $ PZA00029.11   : chr [1:2782] "C/C" "C/C" "C/C" "C/C" ...
##  $ PZA00029.12   : chr [1:2782] "T/T" "T/T" "T/T" "T/T" ...
##  $ PZA00030.11   : chr [1:2782] "?/?" "?/?" "?/?" "?/?" ...
##  $ PZA00031.5    : chr [1:2782] "C/C" "C/C" "C/C" "C/C" ...
##  $ PZA00041.3    : chr [1:2782] "?/?" "?/?" "?/?" "?/?" ...
##  $ PZA00042.2    : chr [1:2782] "?/?" "T/T" "?/?" "?/?" ...
##  $ PZA00042.5    : chr [1:2782] "C/C" "C/C" "C/C" "C/C" ...
##  $ PZA00043.7    : chr [1:2782] "T/T" "T/T" "T/T" "T/T" ...
##  $ PZA00045.1    : chr [1:2782] "G/G" "G/G" "G/G" "G/G" ...
##  $ PZA00047.2    : chr [1:2782] "C/C" "C/C" "C/C" "C/C" ...
##  $ PZA00049.12   : chr [1:2782] "?/?" "?/?" "T/T" "T/T" ...
##  $ PZA00050.9    : chr [1:2782] "A/A" "A/A" "?/?" "?/?" ...
##  $ PZA00051.2    : chr [1:2782] "A/A" "A/A" "A/A" "A/A" ...
##  $ PZA00058.5    : chr [1:2782] "?/?" "?/?" "?/?" "?/?" ...
##  $ PZA00058.6    : chr [1:2782] "T/T" "?/?" "T/T" "T/T" ...
##  $ PZA00060.2    : chr [1:2782] "?/?" "?/?" "C/C" "C/C" ...
##  $ PZA00061.1    : chr [1:2782] "?/?" "?/?" "?/?" "?/?" ...
##  $ PZA00065.2    : chr [1:2782] "C/C" "C/C" "C/C" "C/C" ...
##  $ PZA00069.4    : chr [1:2782] "C/C" "C/C" "C/C" "C/C" ...
##  $ PZA00070.5    : chr [1:2782] "C/C" "C/C" "C/C" "C/C" ...
##  $ PZA00078.2    : chr [1:2782] "C/C" "C/C" "C/C" "C/C" ...
##  $ PZA00079.1    : chr [1:2782] "C/C" "?/?" "C/C" "C/C" ...
##  $ PZA00081.17   : chr [1:2782] "?/?" "T/T" "T/T" "T/T" ...
##  $ PZA00084.2    : chr [1:2782] "C/C" "C/C" "C/C" "C/C" ...
##  $ PZA00084.3    : chr [1:2782] "T/T" "T/T" "T/T" "T/T" ...
##  $ PZA00086.8    : chr [1:2782] "?/?" "?/?" "?/?" "?/?" ...
##  $ PZA00088.3    : chr [1:2782] "G/G" "G/G" "G/G" "G/G" ...
##  $ PZA00090.2    : chr [1:2782] "A/A" "A/A" "?/?" "?/?" ...
##  $ PZA00092.1    : chr [1:2782] "?/?" "?/?" "T/T" "?/?" ...
##  $ PZA00092.5    : chr [1:2782] "?/?" "C/C" "C/C" "?/?" ...
##  $ PZA00093.2    : chr [1:2782] "?/?" "?/?" "?/?" "?/?" ...
##  $ PZA00096.26   : chr [1:2782] "T/T" "T/T" "T/T" "T/T" ...
##  $ PZA00097.13   : chr [1:2782] "G/G" "G/G" "G/G" "G/G" ...
##  $ PZA00098.14   : chr [1:2782] "?/?" "?/?" "C/C" "?/?" ...
##  $ PZA00100.10   : chr [1:2782] "?/?" "?/?" "?/?" "?/?" ...
##  $ PZA00100.12   : chr [1:2782] "T/T" "?/?" "T/T" "T/T" ...
##  $ PZA00100.14   : chr [1:2782] "?/?" "A/A" "A/A" "A/A" ...
##  $ PZA00100.9    : chr [1:2782] "C/C" "C/C" "C/C" "?/?" ...
##  $ PZA00103.20   : chr [1:2782] "A/A" "A/A" "A/A" "A/A" ...
##  $ PZA00106.9    : chr [1:2782] "G/G" "G/G" "G/G" "G/G" ...
##  $ PZA00107.18   : chr [1:2782] "?/?" "?/?" "?/?" "?/?" ...
##  $ PZA00108.12   : chr [1:2782] "?/?" "C/C" "C/C" "C/C" ...
##  $ PZA00108.14   : chr [1:2782] "G/G" "G/G" "G/G" "G/G" ...
##  $ PZA00108.15   : chr [1:2782] "A/A" "A/A" "A/A" "A/A" ...
```

```
##  $ PZA00109.3    : chr [1:2782] "A/A" "A/A" "?/?" "?/?" ...
##  $ PZA00109.5    : chr [1:2782] "A/A" "?/?" "A/A" "?/?" ...
##  $ PZA00111.2    : chr [1:2782] "T/T" "T/T" "T/T" "T/T" ...
##  $ PZA00111.4    : chr [1:2782] "C/C" "?/?" "C/C" "C/C" ...
##  $ PZA00111.5    : chr [1:2782] "?/?" "?/?" "A/A" "A/A" ...
##  $ PZA00111.6    : chr [1:2782] "C/C" "C/C" "C/C" "C/C" ...
##  $ PZA00111.8    : chr [1:2782] "T/T" "T/T" "T/T" "T/T" ...
##  $ PZA00114.3    : chr [1:2782] "C/C" "C/C" "C/C" "C/C" ...
##  $ PZA00116.2    : chr [1:2782] "C/T" "C/T" "C/T" "C/T" ...
##  $ PZA00119.4    : chr [1:2782] "G/G" "G/G" "G/G" "G/G" ...
##  $ PZA00120.4    : chr [1:2782] "G/G" "G/G" "G/G" "G/G" ...
##  $ PZA00123.1    : chr [1:2782] "?/?" "?/?" "?/?" "?/?" ...
##  $ PZA00125.2    : chr [1:2782] "?/?" "?/?" "?/?" "?/?" ...
##  $ PZA00131.14   : chr [1:2782] "C/C" "C/C" "C/C" "C/C" ...
##  $ PZA00132.17   : chr [1:2782] "T/T" "T/T" "T/T" "T/T" ...
##   [list output truncated]
##  - attr(*, "spec")=
##   .. cols(
##   ..   Sample_ID = col_character(),
##   ..   JG_OTU = col_character(),
##   ..   Group = col_character(),
##   ..   abph1.20 = col_character(),
##   ..   abph1.22 = col_character(),
##   ..   ae1.3 = col_character(),
##   ..   ae1.4 = col_character(),
##   ..   ae1.5 = col_character(),
##   ..   an1.4 = col_character(),
##   ..   ba1.6 = col_character(),
##   ..   ba1.9 = col_character(),
##   ..   bt2.5 = col_character(),
##   ..   bt2.7 = col_character(),
##   ..   bt2.8 = col_character(),
##   ..   Fea2.1 = col_character(),
##   ..   Fea2.5 = col_character(),
##   ..   id1.3 = col_character(),
##   ..   lg2.11 = col_character(),
##   ..   lg2.2 = col_character(),
##   ..   pbf1.1 = col_character(),
##   ..   pbf1.2 = col_character(),
##   ..   pbf1.3 = col_character(),
##   ..   pbf1.5 = col_character(),
##   ..   pbf1.6 = col_character(),
##   ..   pbf1.7 = col_character(),
##   ..   pbf1.8 = col_character(),
##   ..   PZA00003.11 = col_character(),
##   ..   PZA00004.2 = col_character(),
##   ..   PZA00005.8 = col_character(),
##   ..   PZA00005.9 = col_character(),
##   ..   PZA00006.13 = col_character(),
##   ..   PZA00006.14 = col_character(),
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##   ..   PZA00013.11 = col_character(),
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##   ..   PZA00111.4 = col_character(),
##   ..   PZA00111.5 = col_character(),
##   ..   PZA00111.6 = col_character(),
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##   ..   PZA00210.6 = col_character(),
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##   ..   PZA00326.19 = col_character(),
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##   ..   PZA00364.6 = col_character(),
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##   ..   PZA00422.6 = col_character(),
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##   ..   PZA01240.2 = col_character(),
##   ..   PZA01420.1 = col_character(),
##   ..   PZA01420.2 = col_character(),
##   ..   PZA01420.3 = col_character(),
##   ..   PZA01474.2 = col_character(),
##   ..   PZA01637.2 = col_character(),
##   ..   PZA01637.3 = col_character(),
##   ..   PZA01637.4 = col_character(),
##   ..   PZA01725.1 = col_character(),
##   ..   PZA01725.2 = col_character(),
##   ..   PZA01782.2 = col_character(),
##   ..   PZA01782.3 = col_character(),
##   ..   PZA01782.4 = col_character(),
##   ..   PZA02789.31 = col_character(),
##   ..   PZA02789.36 = col_character(),
##   ..   PZA02791.6 = col_character(),
```

```
##   ..   PZA02792.16 = col_character(),
##   ..   PZA02792.9 = col_character(),
##   ..   PZA02806.4 = col_character(),
##   ..   PZA02806.9 = col_character(),
##   ..   PZA02807.5 = col_character(),
##   ..   PZA02808.12 = col_character(),
##   ..   PZA02808.16 = col_character(),
##   ..   PZA02819.35 = col_character(),
##   ..   PZA02820.6 = col_character(),
##   ..   PZA02822.2 = col_character(),
##   ..   PZA02824.1 = col_character(),
##   ..   PZA02824.3 = col_character(),
##   ..   PZA02825.8 = col_character(),
##   ..   PZA02831.5 = col_character(),
##   ..   PZA02837.5 = col_character(),
##   ..   PZA02844.1 = col_character(),
##   ..   PZA02850.18 = col_character(),
##   ..   PZA02850.4 = col_character(),
##   ..   PZA02853.10 = col_character(),
##   ..   PZA02853.7 = col_character(),
##   ..   PZA02856.1 = col_character(),
##   ..   PZA02862.3 = col_character(),
##   ..   PZA02865.11 = col_character(),
##   ..   PZA02869.2 = col_character(),
##   ..   PZA02869.8 = col_character(),
##   ..   PZA02872.1 = col_character(),
##   ..   PZA02872.3 = col_character(),
##   ..   PZA02878.12 = col_character(),
##   ..   PZA02888.3 = col_character(),
##   ..   PZA02890.3 = col_character(),
##   ..   PZA02890.4 = col_character(),
##   ..   PZA02890.5 = col_character(),
##   ..   PZA02894.1 = col_character(),
##   ..   PZA02897.12 = col_character(),
##   ..   PZA02906.12 = col_character(),
##   ..   PZA02906.7 = col_character(),
##   ..   PZA02921.9 = col_character(),
##   ..   PZA02923.7 = col_character(),
##   ..   PZA02927.1 = col_character(),
##   ..   PZA02938.5 = col_character(),
##   ..   PZA02939.6 = col_character(),
##   ..   PZA02940.3 = col_character(),
##   ..   PZA02941.3 = col_character(),
##   ..   PZA02941.6 = col_character(),
##   ..   PZA02941.8 = col_character(),
##   ..   PZA02947.2 = col_character(),
##   ..   PZA02948.19 = col_character(),
##   ..   PZA02948.21 = col_character(),
##   ..   PZA02948.22 = col_character(),
##   ..   PZA02949.22 = col_character(),
##   ..   PZA02949.26 = col_character(),
##   ..   PZA02952.10 = col_character(),
##   ..   PZA02954.2 = col_character(),
##   ..   PZA02955.3 = col_character(),
```

```
##   ..   PZA02958.17 = col_character(),
##   ..   PZA02959.7 = col_character(),
##   ..   PZA02961.1 = col_character(),
##   ..   PZA02962.13 = col_character(),
##   ..   PZA02963.5 = col_character(),
##   ..   PZA02966.11 = col_character(),
##   ..   PZA02968.4 = col_character(),
##   ..   PZA02969.11 = col_character(),
##   ..   PZA02970.9 = col_character(),
##   ..   PZA02972.1 = col_character(),
##   ..   PZA02982.5 = col_character(),
##   ..   PZA02982.6 = col_character(),
##   ..   PZA02983.38 = col_character(),
##   ..   PZA02984.7 = col_character(),
##   ..   PZA02988.2 = col_character(),
##   ..   PZA02993.5 = col_character(),
##   ..   PZA02997.16 = col_character(),
##   ..   PZA02997.19 = col_character(),
##   ..   PZA03001.15 = col_character(),
##   ..   PZA03001.18 = col_character(),
##   ..   PZA03001.9 = col_character(),
##   ..   PZA03009.5 = col_character(),
##   ..   PZA03009.6 = col_character(),
##   ..   PZA03009.7 = col_character(),
##   ..   PZA03009.8 = col_character(),
##   ..   PZA03011.6 = col_character(),
##   ..   PZA03012.10 = col_character(),
##   ..   PZA03013.7 = col_character(),
##   ..   PZA03013.8 = col_character(),
##   ..   PZA03014.10 = col_character(),
##   ..   PZA03014.21 = col_character(),
##   ..   PZA03014.24 = col_character(),
##   ..   PZA03017.10 = col_character(),
##   ..   PZA03017.11 = col_character(),
##   ..   PZA03024.16 = col_character(),
##   ..   PZA03024.18 = col_character(),
##   ..   PZA03024.7 = col_character(),
##   ..   PZA03028.5 = col_character(),
##   ..   PZA03032.16 = col_character(),
##   ..   PZA03034.1 = col_character(),
##   ..   PZA03035.5 = col_character(),
##   ..   PZA03037.8 = col_character(),
##   ..   PZA03037.9 = col_character(),
##   ..   PZA03041.8 = col_character(),
##   ..   PZA03042.1 = col_character(),
##   ..   PZA03042.5 = col_character(),
##   ..   PZA03046.2 = col_character(),
##   ..   PZA03046.3 = col_character(),
##   ..   PZA03047.12 = col_character(),
##   ..   PZA03047.20 = col_character(),
##   ..   PZA03047.22 = col_character(),
##   ..   PZA03048.16 = col_character(),
##   ..   PZA03048.17 = col_character(),
##   ..   PZA03049.23 = col_character(),
```

```
##   ..   PZA03051.1 = col_character(),
##   ..   PZA03051.3 = col_character(),
##   ..   PZA03052.15 = col_character(),
##   ..   PZA03054.3 = col_character(),
##   ..   PZA03054.5 = col_character(),
##   ..   PZA03058.17 = col_character(),
##   ..   PZA03062.15 = col_character(),
##   ..   PZA03062.7 = col_character(),
##   ..   PZA03063.17 = col_character(),
##   ..   PZA03063.18 = col_character(),
##   ..   PZA03064.6 = col_character(),
##   ..   PZA03067.17 = col_character(),
##   ..   PZA03067.20 = col_character(),
##   ..   PZA03068.11 = col_character(),
##   ..   PZA03068.13 = col_character(),
##   ..   PZA03069.6 = col_character(),
##   ..   PZA03073.23 = col_character(),
##   ..   PZA03073.24 = col_character(),
##   ..   PZA03074.24 = col_character(),
##   ..   PZA03078.33 = col_character(),
##   ..   PZA03081.1 = col_character(),
##   ..   PZA03081.10 = col_character(),
##   ..   PZA03081.11 = col_character(),
##   ..   PZA03081.13 = col_character(),
##   ..   PZA03081.6 = col_character(),
##   ..   PZA03083.7 = col_character(),
##   ..   PZA03089.12 = col_character(),
##   ..   PZA03090.31 = col_character(),
##   ..   PZA03092.7 = col_character(),
##   ..   PZA03094.18 = col_character(),
##   ..   PZA03094.6 = col_character(),
##   ..   PZA03095.1 = col_character(),
##   ..   PZA03095.2 = col_character(),
##   ..   PZA03095.3 = col_character(),
##   ..   PZA03097.4 = col_character(),
##   ..   PZA03097.7 = col_character(),
##   ..   PZA03097.9 = col_character(),
##   ..   PZA03102.10 = col_character(),
##   ..   PZA03102.2 = col_character(),
##   ..   PZA03102.9 = col_character(),
##   ..   PZA03137.1 = col_character(),
##   ..   PZA03172.2 = col_character(),
##   ..   PZA03223.3 = col_character(),
##   ..   PZA03258.2 = col_character(),
##   ..   PZA03283.2 = col_character(),
##   ..   PZA03284.3 = col_character(),
##   ..   PZA03290.1 = col_character(),
##   ..   PZA03290.2 = col_character(),
##   ..   PZA03295.4 = col_character(),
##   ..   PZA03296.6 = col_character(),
##   ..   PZA03296.7 = col_character(),
##   ..   PZA03298.1 = col_character(),
##   ..   PZA03298.2 = col_character(),
##   ..   PZA03301.2 = col_character(),
```

```
##   ..   PZA03301.4 = col_character(),
##   ..   PZA03302.1 = col_character(),
##   ..   PZA03305.6 = col_character(),
##   ..   PZA03305.7 = col_character(),
##   ..   PZA03311.2 = col_character(),
##   ..   PZA03311.3 = col_character(),
##   ..   PZA03311.4 = col_character(),
##   ..   PZA03311.5 = col_character(),
##   ..   PZA03312.1 = col_character(),
##   ..   PZA03312.2 = col_character(),
##   ..   PZA03316.2 = col_character(),
##   ..   PZA03317.1 = col_character(),
##   ..   PZA03319.3 = col_character(),
##   ..   PZA03319.4 = col_character(),
##   ..   PZA03320.3 = col_character(),
##   ..   PZA03320.4 = col_character(),
##   ..   PZA03328.5 = col_character(),
##   ..   PZA03329.1 = col_character(),
##   ..   PZA03329.2 = col_character(),
##   ..   PZA03333.3 = col_character(),
##   ..   PZA03335.2 = col_character(),
##   ..   PZA03335.3 = col_character(),
##   ..   PZA03337.1 = col_character(),
##   ..   PZA03338.5 = col_character(),
##   ..   PZA03340.2 = col_character(),
##   ..   PZA03342.2 = col_character(),
##   ..   PZA03344.4 = col_character(),
##   ..   PZA03344.5 = col_character(),
##   ..   PZA03344.6 = col_character(),
##   ..   PZA03345.1 = col_character(),
##   ..   PZA03345.2 = col_character(),
##   ..   PZA03345.4 = col_character(),
##   ..   PZA03347.1 = col_character(),
##   ..   PZA03348.1 = col_character(),
##   ..   PZA03349.1 = col_character(),
##   ..   PZA03349.9 = col_character(),
##   ..   PZA03767.1 = col_character(),
##   ..   PZA03767.4 = col_character(),
##   ..   PZA03767.5 = col_character(),
##   ..   PZA03773.2 = col_character(),
##   ..   PZA03773.3 = col_character(),
##   ..   PZA03774.1 = col_character(),
##   ..   PZA03774.10 = col_character(),
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##   ..   PZA03774.4 = col_character(),
##   ..   PZA03774.5 = col_character(),
##   ..   PZA03774.6 = col_character(),
##   ..   PZA03774.8 = col_character(),
##   ..   PZA03774.9 = col_character(),
##   ..   PZA03775.1 = col_character(),
##   ..   PZA03775.11 = col_character(),
##   ..   PZA03775.2 = col_character(),
##   ..   PZA03775.3 = col_character(),
##   ..   PZA03775.4 = col_character(),
```

```
##   ..   PZA03775.6 = col_character(),
##   ..   PZA03775.7 = col_character(),
##   ..   PZA03775.8 = col_character(),
##   ..   PZA03775.9 = col_character(),
##   ..   PZA03781.1 = col_character(),
##   ..   PZA03781.2 = col_character(),
##   ..   PZA03781.3 = col_character(),
##   ..   PZA03781.4 = col_character(),
##   ..   PZA03781.5 = col_character(),
##   ..   PZA03781.6 = col_character(),
##   ..   PZA03781.7 = col_character(),
##   ..   PZA03781.8 = col_character(),
##   ..   PZA03782.1 = col_character(),
##   ..   PZA03782.3 = col_character(),
##   ..   PZA03786.1 = col_character(),
##   ..   PZA03786.2 = col_character(),
##   ..   PZA03789.1 = col_character(),
##   ..   PZA03789.2 = col_character(),
##   ..   PZA03789.4 = col_character(),
##   ..   PZB00011.4 = col_character(),
##   ..   PZB00011.5 = col_character(),
##   ..   PZB00041.2 = col_character(),
##   ..   PZB00041.4 = col_character(),
##   ..   PZB00049.2 = col_character(),
##   ..   PZB00049.4 = col_character(),
##   ..   PZB00049.7 = col_character(),
##   ..   PZB00055.1 = col_character(),
##   ..   PZB00060.4 = col_character(),
##   ..   PZB00062.6 = col_character(),
##   ..   PZB00062.7 = col_character(),
##   ..   PZB00062.8 = col_character(),
##   ..   PZB00067.2 = col_character(),
##   ..   PZB00067.3 = col_character(),
##   ..   PZB00067.4 = col_character(),
##   ..   PZB00067.5 = col_character(),
##   ..   PZB00078.1 = col_character(),
##   ..   PZB00081.2 = col_character(),
##   ..   PZB00081.4 = col_character(),
##   ..   PZB00081.5 = col_character(),
##   ..   PZB00081.7 = col_character(),
##   ..   PZB00092.1 = col_character(),
##   ..   PZB00092.4 = col_character(),
##   ..   PZB00093.3 = col_character(),
##   ..   PZB00093.4 = col_character(),
##   ..   PZB00093.6 = col_character(),
##   ..   PZB00096.2 = col_character(),
##   ..   PZB00096.3 = col_character(),
##   ..   PZB00136.3 = col_character(),
##   ..   PZB00140.1 = col_character(),
##   ..   PZB00145.2 = col_character(),
##   ..   PZB00149.2 = col_character(),
##   ..   PZB00149.4 = col_character(),
##   ..   PZB00153.1 = col_character(),
##   ..   PZB00153.2 = col_character(),
```

```
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##   ..   PZB00153.5 = col_character(),
##   ..   PZB00160.1 = col_character(),
##   ..   PZB00160.2 = col_character(),
##   ..   PZB00160.4 = col_character(),
##   ..   PZB00165.2 = col_character(),
##   ..   PZB00165.6 = col_character(),
##   ..   PZB00169.4 = col_character(),
##   ..   PZB00169.6 = col_character(),
##   ..   PZB00175.1 = col_character(),
##   ..   PZB00175.2 = col_character(),
##   ..   PZB00175.3 = col_character(),
##   ..   PZB00175.4 = col_character(),
##   ..   PZB00175.5 = col_character(),
##   ..   PZB00180.1 = col_character(),
##   ..   PZB00180.2 = col_character(),
##   ..   PZB00183.3 = col_character(),
##   ..   PZB00188.6 = col_character(),
##   ..   PZB00207.3 = col_character(),
##   ..   PZB00221.3 = col_character(),
##   ..   PZB00221.8 = col_character(),
##   ..   PZB00229.3 = col_character(),
##   ..   PZB00232.1 = col_character(),
##   ..   PZB00232.2 = col_character(),
##   ..   PZB00232.4 = col_character(),
##   ..   PZB00232.5 = col_character(),
##   ..   PZB00379.3 = col_character(),
##   ..   PZB00379.4 = col_character(),
##   ..   PZB00379.5 = col_character(),
##   ..   PZB00393.7 = col_character(),
##   ..   PZB00409.3 = col_character(),
##   ..   PZB00416.2 = col_character(),
##   ..   PZB00416.5 = col_character(),
##   ..   PZB00454.2 = col_character(),
##   ..   PZB00454.3 = col_character(),
##   ..   PZB00454.4 = col_character(),
##   ..   PZB00454.5 = col_character(),
##   ..   PZB00498.2 = col_character(),
##   ..   PZB00498.4 = col_character(),
##   ..   PZB00598.1 = col_character(),
##   ..   PZB00598.2 = col_character(),
##   ..   PZB00603.3 = col_character(),
##   ..   PZB00603.4 = col_character(),
##   ..   PZB00603.5 = col_character(),
##   ..   PZB00607.2 = col_character(),
##   ..   PZB00761.1 = col_character(),
##   ..   PZB00761.2 = col_character(),
##   ..   PZB00849.2 = col_character(),
##   ..   PZB00849.3 = col_character(),
##   ..   PZB00849.4 = col_character(),
##   ..   PZB00859.1 = col_character(),
##   ..   PZB01109.2 = col_character(),
##   ..   PZB01109.3 = col_character(),
##   ..   PZB01110.1 = col_character(),
```

```
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##   ..   PZB01110.3 = col_character(),
##   ..   PZB01111.6 = col_character(),
##   ..   PZB01111.7 = col_character(),
##   ..   PZB01111.8 = col_character(),
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##   ..   PZB01112.4 = col_character(),
##   ..   PZB01112.5 = col_character(),
##   ..   PZB01112.6 = col_character(),
##   ..   PZB01113.4 = col_character(),
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##   ..   PZB01114.3 = col_character(),
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##   ..   PZB01115.6 = col_character(),
##   ..   PZB01116.2 = col_character(),
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##   ..   PZB01222.1 = col_character(),
##   ..   PZB01222.3 = col_character(),
##   ..   PZB01223.3 = col_character(),
##   ..   PZB01223.4 = col_character(),
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##   ..   PZB01225.2 = col_character(),
##   ..   PZB01225.4 = col_character(),
##   ..   PZB01228.1 = col_character(),
##   ..   PZB01228.3 = col_character(),
##   ..   PZB01228.4 = col_character(),
##   ..   PZB01233.2 = col_character(),
##   ..   PZB01233.3 = col_character(),
##   ..   PZB01238.5 = col_character(),
##   ..   PZB01238.6 = col_character(),
##   ..   PZB01427.1 = col_character(),
##   ..   PZB01427.3 = col_character(),
##   ..   PZB01463.2 = col_character(),
##   ..   PZB01463.3 = col_character(),
##   ..   PZB01463.4 = col_character(),
##   ..   PZD00003.1 = col_character(),
##   ..   PZD00003.3 = col_character(),
##   ..   PZD00007.1 = col_character(),
##   ..   PZD00008.3 = col_character(),
##   ..   PZD00011.1 = col_character(),
##   ..   PZD00011.3 = col_character(),
##   ..   PZD00011.4 = col_character(),
##   ..   PZD00012.1 = col_character(),
##   ..   PZD00012.2 = col_character(),
##   ..   PZD00012.3 = col_character(),
##   ..   PZD00012.4 = col_character(),
##   ..   PZD00012.5 = col_character(),
##   ..   PZD00013.3 = col_character(),
##   ..   PZD00013.4 = col_character(),
##   ..   PZD00014.3 = col_character(),
##   ..   PZD00017.1 = col_character(),
##   ..   PZD00019.1 = col_character(),
```

```
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##   ..   PZD00020.3 = col_character(),
##   ..   PZD00020.4 = col_character(),
##   ..   PZD00020.6 = col_character(),
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##   ..   PZD00021.4 = col_character(),
##   ..   PZD00021.5 = col_character(),
##   ..   PZD00022.1 = col_character(),
##   ..   PZD00022.3 = col_character(),
##   ..   PZD00022.4 = col_character(),
##   ..   PZD00024.2 = col_character(),
##   ..   PZD00025.1 = col_character(),
##   ..   PZD00025.2 = col_character(),
##   ..   PZD00030.1 = col_character(),
##   ..   PZD00030.4 = col_character(),
##   ..   PZD00030.5 = col_character(),
##   ..   PZD00030.6 = col_character(),
##   ..   PZD00034.3 = col_character(),
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##   ..   PZD00043.3 = col_character(),
##   ..   PZD00043.4 = col_character(),
##   ..   PZD00044.2 = col_character(),
##   ..   PZD00044.3 = col_character(),
##   ..   PZD00044.4 = col_character(),
##   ..   PZD00045.1 = col_character(),
##   ..   PZD00045.2 = col_character(),
##   ..   PZD00045.3 = col_character(),
##   ..   PZD00045.4 = col_character(),
##   ..   PZD00049.3 = col_character(),
##   ..   PZD00049.4 = col_character(),
##   ..   PZD00049.5 = col_character(),
##   ..   PZD00051.1 = col_character(),
##   ..   PZD00052.3 = col_character(),
##   ..   PZD00052.4 = col_character(),
##   ..   PZD00062.2 = col_character(),
##   ..   PZD00066.1 = col_character(),
##   ..   PZD00067.1 = col_character(),
##   ..   PZD00067.2 = col_character(),
##   ..   PZD00067.3 = col_character(),
##   ..   PZD00068.1 = col_character(),
##   ..   PZD00069.2 = col_character(),
##   ..   PZD00069.3 = col_character(),
##   ..   PZD00069.4 = col_character(),
##   ..   PZD00069.5 = col_character(),
##   ..   PZD00073.1 = col_character(),
##   ..   PZD00073.2 = col_character(),
##   ..   PZD00073.6 = col_character(),
##   ..   PZD00074.1 = col_character(),
##   ..   PZD00075.1 = col_character(),
##   ..   PZD00075.2 = col_character(),
##   ..   PZD00076.1 = col_character(),
##   ..   PZD00076.2 = col_character(),
##   ..   PZD00076.4 = col_character(),
```

```
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##   ..   PZD00077.5 = col_character(),
##   ..   PZD00077.7 = col_character(),
##   ..   PZD00077.8 = col_character(),
##   ..   PZD00078.2 = col_character(),
##   ..   Ra2_ORF.1 = col_character(),
##   ..   Ra2_ORF.2 = col_character(),
##   ..   Ra2_ORF.4 = col_character(),
##   ..   Ra2_promoter.1 = col_character(),
##   ..   Ra2_promoter.2 = col_character(),
##   ..   Ra2_promoter.3 = col_character(),
##   ..   sh2.5 = col_character(),
##   ..   sh2.6 = col_character(),
##   ..   sh2.7 = col_character(),
##   ..   sh2.9 = col_character(),
##   ..   su1.4 = col_character(),
##   ..   su1.5 = col_character(),
##   ..   su1.7 = col_character(),
##   ..   tb1.17 = col_character(),
##   ..   tb1.18 = col_character(),
##   ..   tb1.19 = col_character(),
##   ..   tb1.5 = col_character(),
##   ..   te1.3 = col_character(),
##   ..   te1.4 = col_character(),
##   ..   zagl1.1 = col_character(),
##   ..   zagl1.6 = col_character(),
##   ..   zap1.2 = col_character(),
##   ..   zen1.1 = col_character(),
##   ..   zen1.2 = col_character(),
##   ..   zen1.4 = col_character(),
##   ..   zfl2.6 = col_character(),
##   ..   zmm3.4 = col_character()
##   .. )
##  - attr(*, "problems")=<externalptr>
```

```r
head(snp)
```

```
## # A tibble: 6 x 15
##   SNP_ID   cdv_marker_id Chromosome Position  alt_pos mult_positions amplicon
##   <chr>            <dbl> <chr>      <chr>     <chr>   <chr>          <chr>
## 1 abph1.20          5976 2          27403404  <NA>    <NA>           abph1
## 2 abph1.22          5978 2          27403892  <NA>    <NA>           abph1
## 3 ae1.3             6605 5          167889790 <NA>    <NA>           ae1
## 4 ae1.4             6606 5          167889682 <NA>    <NA>           ae1
## 5 ae1.5             6607 5          167889821 <NA>    <NA>           ae1
## 6 an1.4             5982 1          240498509 <NA>    <NA>           an1
## # i 8 more variables: cdv_map_feature.name <chr>, gene <chr>,
## #   `candidate/random` <chr>, Genaissance_daa_id <dbl>, Sequenom_daa_id <dbl>,
## #   count_amplicons <dbl>, count_cmf <dbl>, count_gene <dbl>
```

```r
ncol(snp)
```

```
## [1] 15
```

```r
nrow(snp)
```

```
## [1] 983
```

```r
dim(snp)
```

```
## [1] 983  15
```

```r
colnames(snp)
```

```
##  [1] "SNP_ID"               "cdv_marker_id"        "Chromosome"
##  [4] "Position"             "alt_pos"              "mult_positions"
##  [7] "amplicon"             "cdv_map_feature.name" "gene"
## [10] "candidate/random"     "Genaissance_daa_id"   "Sequenom_daa_id"
## [13] "count_amplicons"      "count_cmf"            "count_gene"
```

```r
str(snp)
```

```
## spc_tbl_ [983 x 15] (S3: spec_tbl_df/tbl_df/tbl/data.frame)
##  $ SNP_ID              : chr [1:983] "abph1.20" "abph1.22" "ae1.3" "ae1.4" ...
##  $ cdv_marker_id       : num [1:983] 5976 5978 6605 6606 6607 ...
##  $ Chromosome          : chr [1:983] "2" "2" "5" "5" ...
##  $ Position            : chr [1:983] "27403404" "27403892" "167889790" "167889682" ...
##  $ alt_pos             : chr [1:983] NA NA NA NA ...
##  $ mult_positions      : chr [1:983] NA NA NA NA ...
##  $ amplicon            : chr [1:983] "abph1" "abph1" "ae1" "ae1" ...
##  $ cdv_map_feature.name: chr [1:983] "AB042260" "AB042260" "ae1" "ae1" ...
##  $ gene                : chr [1:983] "abph1" "abph1" "ae1" "ae1" ...
##  $ candidate/random    : chr [1:983] "candidate" "candidate" "candidate" "candidate" ...
##  $ Genaissance_daa_id  : num [1:983] 8393 8394 8395 8396 8397 ...
##  $ Sequenom_daa_id     : num [1:983] 10474 10475 10477 10478 10479 ...
##  $ count_amplicons     : num [1:983] 1 0 1 0 0 1 1 0 1 0 ...
##  $ count_cmf           : num [1:983] 1 0 1 0 0 1 0 0 1 0 ...
##  $ count_gene          : num [1:983] 1 0 1 0 0 1 1 0 1 0 ...
##  - attr(*, "spec")=
##   .. cols(
##   ..   SNP_ID = col_character(),
##   ..   cdv_marker_id = col_double(),
##   ..   Chromosome = col_character(),
##   ..   Position = col_character(),
##   ..   alt_pos = col_character(),
##   ..   mult_positions = col_character(),
##   ..   amplicon = col_character(),
##   ..   cdv_map_feature.name = col_character(),
##   ..   gene = col_character(),
##   ..   `candidate/random` = col_character(),
##   ..   Genaissance_daa_id = col_double(),
##   ..   Sequenom_daa_id = col_double(),
##   ..   count_amplicons = col_double(),
##   ..   count_cmf = col_double(),
##   ..   count_gene = col_double()
##   .. )
##  - attr(*, "problems")=<externalptr>
```

fang *We have 986 columns, and 2782 rows. Columns include Sample_ID, individual genotypes, and metadata.* **snp** *We have 15 columns, and 983 rows. These rows are the SNPs. The columns includes SNP Id, chromosome and nucleotide position*

---

2. Data Processing

```
# Transpose fang_et_el data
fang_t <- as.data.frame(fang)
fang_t$SNP_ID <- rownames(fang_t)

# make the genotypes data a "long format" for easier manipulation.
fang_long <- fang_t %>%
  pivot_longer(
    cols = 4:ncol(fang_t),
    names_to = "SNP_ID",
    values_to = "Genotype"
  )

head(fang_long)
```

Transposing and merging

```
## # A tibble: 6 x 5
##   Sample_ID JG_OTU   Group SNP_ID   Genotype
##   <chr>     <chr>    <chr> <chr>    <chr>
## 1 SL-15     T-aust-1 TRIPS abph1.20 ?/?
## 2 SL-15     T-aust-1 TRIPS abph1.22 ?/?
## 3 SL-15     T-aust-1 TRIPS ae1.3    T/T
## 4 SL-15     T-aust-1 TRIPS ae1.4    G/G
## 5 SL-15     T-aust-1 TRIPS ae1.5    T/T
## 6 SL-15     T-aust-1 TRIPS an1.4    C/C
```

```
dim(fang_long)
```

```
## [1] 2737488       5
```

```
# join genotypes data with the snp positions by SNP_ID
merged_data <- inner_join(fang_long, snp, by = "SNP_ID")
dim(merged_data)
```

```
## [1] 2734706      19
```

```
head(merged_data)
```

```
## # A tibble: 6 x 19
##   Sample_ID JG_OTU   Group SNP_ID   Genotype cdv_marker_id Chromosome Position
##   <chr>     <chr>    <chr> <chr>    <chr>            <dbl> <chr>      <chr>
```

```
## 1 SL-15     T-aust-1 TRIPS abph1.20 ?/?               5976 2          27403404
## 2 SL-15     T-aust-1 TRIPS abph1.22 ?/?               5978 2          27403892
## 3 SL-15     T-aust-1 TRIPS ae1.3    T/T               6605 5          167889790
## 4 SL-15     T-aust-1 TRIPS ae1.4    G/G               6606 5          167889682
## 5 SL-15     T-aust-1 TRIPS ae1.5    T/T               6607 5          167889821
## 6 SL-15     T-aust-1 TRIPS an1.4    C/C               5982 1          240498509
## # i 11 more variables: alt_pos <chr>, mult_positions <chr>, amplicon <chr>,
## #   cdv_map_feature.name <chr>, gene <chr>, `candidate/random` <chr>,
## #   Genaissance_daa_id <dbl>, Sequenom_daa_id <dbl>, count_amplicons <dbl>,
## #   count_cmf <dbl>, count_gene <dbl>
```

```
# Convert back to "wide format" with `SNP_ID`, `Chromosome`, `Position`, and sample genotypes as column
merged_wide <- merged_data %>%
  select(SNP_ID, Chromosome, Position, Sample_ID, Genotype) %>%
  pivot_wider(
    names_from = Sample_ID,
    values_from = Genotype
  )

dim(merged_wide)
```

```
## [1]  983 2785
```

```
head(merged_wide)
```

```
## # A tibble: 6 x 2,785
##   SNP_ID   Chromosome Position  `SL-15` `SL-16` `SL-11` `SL-12` `SL-18` `SL-2`
##   <chr>    <chr>      <chr>     <chr>   <chr>   <chr>   <chr>   <chr>   <chr>
## 1 abph1.20 2          27403404  ?/?     ?/?     ?/?     ?/?     ?/?     ?/?
## 2 abph1.22 2          27403892  ?/?     ?/?     ?/?     ?/?     ?/?     ?/?
## 3 ae1.3    5          167889790 T/T     T/T     T/T     T/T     T/T     T/T
## 4 ae1.4    5          167889682 G/G     ?/?     G/G     G/G     G/G     G/G
## 5 ae1.5    5          167889821 T/T     T/T     T/T     T/T     T/T     T/T
## 6 an1.4    1          240498509 C/C     C/C     ?/?     C/C     C/C     C/C
## # i 2,776 more variables: `SL-4` <chr>, `SL-6` <chr>, `SL-7` <chr>,
## #   `SL-5` <chr>, `SL-3` <chr>, `SL-8` <chr>, `SL-24` <chr>, `SL-22` <chr>,
## #   `SL-14` <chr>, `SL-23` <chr>, `SL-20` <chr>, `SL-19` <chr>, `SL-17` <chr>,
## #   `SL-21` <chr>, `SL-9` <chr>, `SL-10` <chr>, ZDP_1280a <chr>,
## #   ZDP_1312a <chr>, ZDP_1454b <chr>, `JVLJ-690-1` <chr>, `JVLJ-690-19` <chr>,
## #   `JVLJ-690-3` <chr>, `JVLJ-690-5` <chr>, `JVLJ-690-7` <chr>,
## #   `JVLJ-690-9` <chr>, `JVLJ-690-11` <chr>, `JVLJ-690-13` <chr>, ...
```

```
# Split datasets into "maize" and "teosinte" groups
maize_samples <- fang_t %>%
  filter(Group %in% c("ZMMIL","ZMMLR","ZMMMR")) %>%
  pull(Sample_ID)

teosinte_samples <- fang_t %>%
  filter(Group %in% c("ZMPBA","ZMPIL","ZMPJA")) %>%
  pull(Sample_ID)
```

```r
# subset
maize <- merged_wide %>%
  select(SNP_ID, Chromosome, Position, all_of(maize_samples))

teosinte <- merged_wide %>%
  select(SNP_ID, Chromosome, Position, all_of(teosinte_samples))
```

Split maize and teosinte groups

```r
# Generate 40 files
write_chr_files <- function(data, group_name){

lapply(1:10, function(chr){

  chr_data <- data %>%
  filter(Chromosome == chr)

# Increasing position
inc <- chr_data %>%
  arrange(Position) %>%
  mutate(across(4:ncol(.), ~replace_na(., "?")))

write.table(
  inc,
  paste0("results/", group_name, "_chr", chr, "_increasing.txt"),
  sep="\t",
  row.names=FALSE,
  quote=FALSE
  )

# Decreasing position
dec <- chr_data %>%
  arrange(desc(Position)) %>%
  mutate(across(4:ncol(.), ~replace_na(., "-")))

write.table(
  dec,
  paste0("results/", group_name, "_chr", chr, "_decreasing.txt"),
  sep="\t",
  row.names=FALSE,
  quote=FALSE
  )

  })

}

write_chr_files(maize, "maize")
```

Geneerating chromosomes files

```
## [[1]]
## NULL
##
## [[2]]
## NULL
##
## [[3]]
## NULL
##
## [[4]]
## NULL
##
## [[5]]
## NULL
##
## [[6]]
## NULL
##
## [[7]]
## NULL
##
## [[8]]
## NULL
##
## [[9]]
## NULL
##
## [[10]]
## NULL
```

```r
write_chr_files(teosinte, "teosinte")
```

```
## [[1]]
## NULL
##
## [[2]]
## NULL
##
## [[3]]
## NULL
##
## [[4]]
## NULL
##
## [[5]]
## NULL
##
## [[6]]
## NULL
##
## [[7]]
## NULL
##
## [[8]]
```

```
## NULL
##
## [[9]]
## NULL
##
## [[10]]
## NULL
```

```r
colnames(maize)[1:3]
```

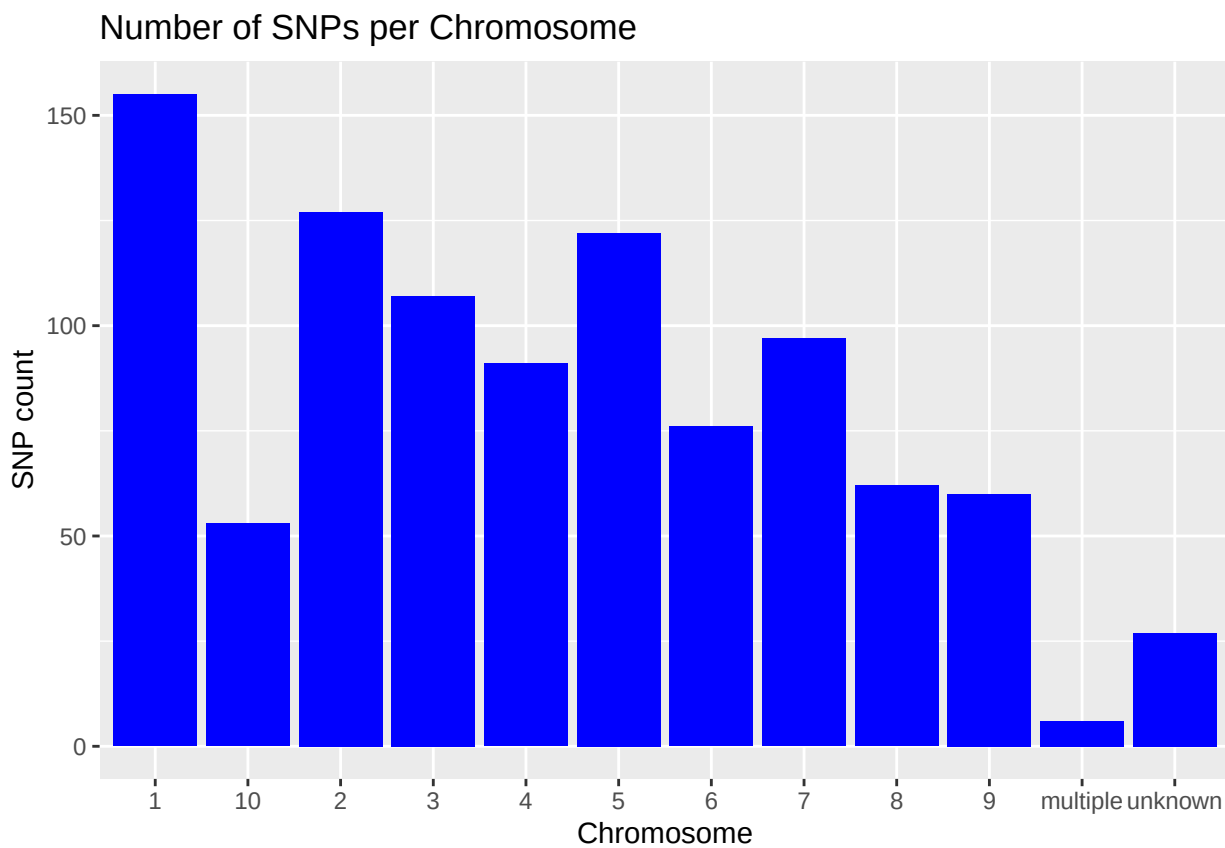
```
## [1] "SNP_ID"     "Chromosome" "Position"
```

Note that these files were saved in my folder called "results", so if you want to regenerate the files into your working directory, replace the "results" in the code with whatever the folder you would like to save them in your current working directory. Or create a folder called "results" first. Thanks.

Part II: Visualization

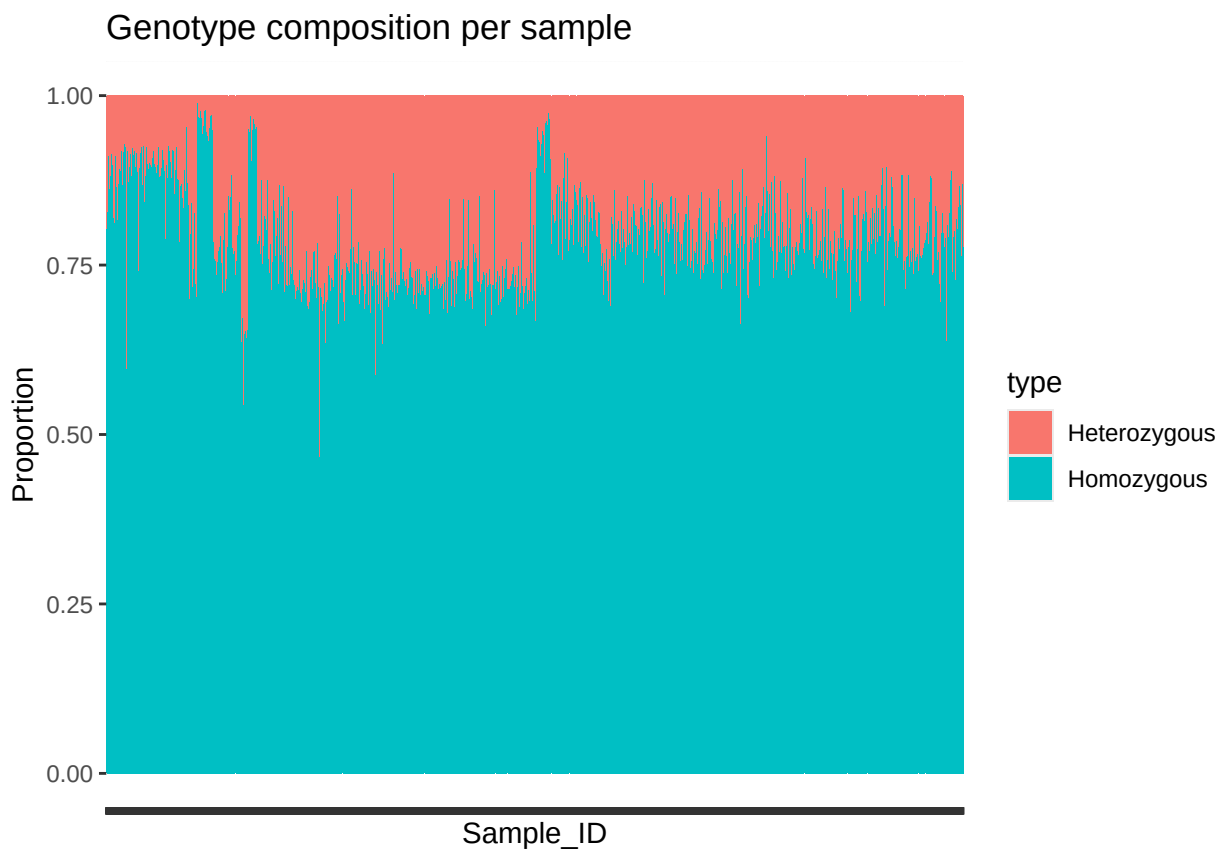
SNPs per chromosome

```r
# SNPs per chromosome
merged_wide %>%
  count(Chromosome) %>%
  ggplot(aes(x=factor(Chromosome), y=n)) +
  geom_bar(stat="identity", fill="blue") +
  labs(
    title="Number of SNPs per Chromosome",
    x="Chromosome",
    y="SNP count"
  )
```



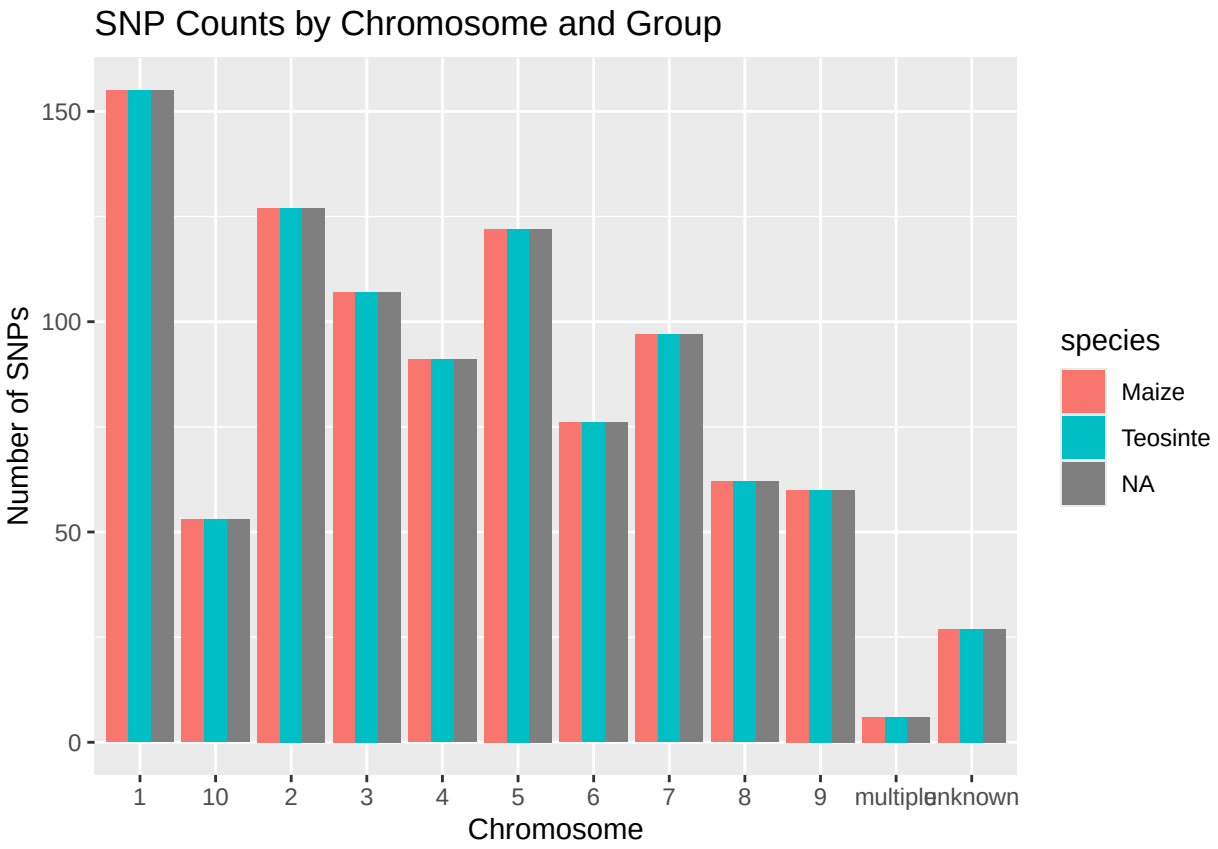
Missing data and amount of heterozygosity

```
# Missing data and amount of heterozygosity
geno_long <- merged_data %>%
  mutate(
    type = case_when(
      Genotype %in% c("A/A","C/C","G/G","T/T") ~ "Homozygous",
      is.na(Genotype) ~ "Missing",
      TRUE ~ "Heterozygous"
    )
  )


ggplot(geno_long, aes(x=Sample_ID, fill=type)) +
  geom_bar(position="fill") +
  labs(
    y="Proportion",
    title="Genotype composition per sample"
  ) +
  theme(axis.text.x = element_blank())
```



SNP counts by chromosome in maize vs teosinte

```
# SNP counts by chromosome in maize vs teosinte
geno_long <- merged_data %>%
  mutate(
    species = case_when(
      Group %in% c("ZMMIL","ZMMLR","ZMMMR") ~ "Maize",
      Group %in% c("ZMPBA","ZMPIL","ZMPJA") ~ "Teosinte"
    )
  )

geno_long %>%
  distinct(SNP_ID, Chromosome, species) %>%
  count(Chromosome, species) %>%
  ggplot(aes(x = factor(Chromosome), y = n, fill = species)) +
  geom_bar(stat = "identity", position = "dodge") +
  labs(
    title = "SNP Counts by Chromosome and Group",
    x = "Chromosome",
    y = "Number of SNPs"
  )
```


SNP Counts by Chromosome and Group
Number of SNPs
150
100
50
0
1
10
2
3
4
5
6
7
8
9
multiple
unknown
Chromosome
species
Maize
Teosinte
NA